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MSc Data Science

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MULTICLASS CLASSIFICATION OF HUMAN ACTIVITIES USING SMARTPHONE SENSOR DATA

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MSc Final Project Declaration

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1. Abstract

This report presents a comprehensive evaluation of machine learning approaches for multiclass classification of human activities using smartphone accelerometer and gyroscope sensor data. Using the UCI Human Activity Recognition (HAR) dataset comprising 10,299 samples with 561 engineered time- and frequency-domain features across six activity classes, we implement and systematically compare three classification algorithms: Logistic Regression, Random Forest, and Support Vector Machine (SVM) with RBF kernel.

Key methodological contributions include: (1) standardised preprocessing using scikit-learn Pipelines to prevent data leakage, (2) comprehensive exploratory data analysis including PCA and t-SNE visualisations revealing class separability patterns, (3) hyperparameter tuning via GridSearchCV and RandomizedSearchCV with 5-fold stratified cross-validation, and (4) detailed analysis of why tuning produces modest improvements on expertly-engineered features.

Results demonstrate that all three models achieve high classification accuracy: Logistic Regression (95.45%), Random Forest (92.74%), and SVM (95.18%). After hyperparameter tuning, SVM achieves the best performance at 95.42% accuracy ($C=10$, $\gamma='scale'$), representing a +0.24% improvement. The modest gains from tuning are attributed to the high-quality feature engineering in the UCI HAR dataset, near-optimal default parameters in scikit-learn, and near-ceiling baseline performance. The primary driver of classification success is the expertly-crafted 561-dimensional feature space rather than hyperparameter optimisation.

2. Introduction

2.1 Background and Motivation

Human Activity Recognition (HAR) has emerged as a critical research area in ubiquitous computing, with applications spanning healthcare monitoring, elderly care, fitness tracking, smart home automation, and rehabilitation. The proliferation of smartphones equipped with inertial measurement units (accelerometers and gyroscopes) has made it feasible to collect high-frequency motion data from users in naturalistic settings without requiring dedicated wearable sensors.

Accurate activity classification enables intelligent context-aware systems. For example, in fall detection for elderly care, distinguishing between sitting, standing, and dynamic activities such as walking or descending stairs is essential for triggering timely alerts. In rehabilitation, tracking whether patients perform prescribed exercises (walking, climbing stairs) provides objective progress monitoring. These real-world applications demand robust, high-accuracy classifiers that can generalise across individuals and conditions.

The UCI HAR dataset (Anguita et al., 2013) provides a well-controlled benchmark for evaluating activity recognition algorithms. Collected from 30 volunteers wearing a Samsung Galaxy S II smartphone at the waist, the dataset captures tri-axial acceleration and angular velocity at 50 Hz during six daily activities: WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, and LAYING. The raw signals have been preprocessed into 561 expertly-engineered features using established signal processing techniques, making it ideal for evaluating classical machine learning approaches.

2.2 Research Question and Objectives

This project addresses the primary research question: Can machine learning models accurately classify human activities into six categories using accelerometer and gyroscope sensor data collected from smartphones? The specific objectives are:

- Implement and evaluate three diverse classifiers (Logistic Regression, Random Forest, SVM) representing different algorithmic paradigms.
- Perform comprehensive exploratory data analysis to understand class distributions, feature characteristics, and class separability through dimensionality reduction.
- Apply hyperparameter tuning with cross-validation to optimise each model and assess whether tuning yields meaningful improvements.
- Analyse and explain why tuning may not produce large gains on certain datasets, providing insight into the relationship between feature quality and model performance.

2.3 Report Structure

Section 3 reviews relevant literature on human activity recognition methods. Section 4 details the methodology including dataset description, exploratory analysis, model specifications, and evaluation metrics. Section 5 presents results with confusion matrices and tuning analysis. Section 6 discusses findings, limitations, and ethical considerations. Section 7 concludes with key contributions and future directions.

3. Literature Review

3.1 Traditional Approaches to HAR

Early approaches to human activity recognition relied on threshold-based heuristics applied to raw accelerometer readings (Ravi et al., 2005). These rule-based systems achieved reasonable accuracy for simple binary classifications (active vs. sedentary) but failed to distinguish between nuanced activities such as walking upstairs versus walking downstairs. The transition to feature-engineered machine learning pipelines, pioneered by Anguita et al. (2012), represented a paradigm shift by extracting statistical and spectral features from fixed-length sliding windows of sensor signals.

3.2 Machine Learning for Activity Recognition

Anguita et al. (2013) demonstrated that multiclass SVM classifiers achieve over 96% accuracy on the UCI HAR dataset when using their 561-feature representation. Their work established the feature engineering pipeline (noise filtering, gravitational/body acceleration separation, time-domain statistics, FFT-based frequency features) that forms the foundation of the dataset used in this project. Subsequent studies have applied Random Forests (Ignatov, 2018), gradient boosting methods, and deep learning approaches (Hammerla et al., 2016), though the classical ML methods remain competitive on this specific dataset due to the high quality of the hand-crafted features.

Logistic Regression, despite its linear nature, has been shown to perform surprisingly well on high-dimensional feature spaces where classes are approximately linearly separable in the transformed space (Hosmer et al., 2013). For the HAR dataset, the 561 features include non-linear transformations (energy, entropy, correlation) that effectively project the original signal space into a more separable representation.

Ensemble methods such as Random Forests aggregate predictions from multiple decorrelated decision trees, reducing variance and improving generalisation. However, axis-aligned splits can be sub-optimal in high-dimensional spaces where class boundaries are oblique rather than axis-parallel. Gradient boosting frameworks such as XGBoost and LightGBM have shown marginal improvements over Random Forests on the UCI HAR benchmark, but the additional computational cost and hyperparameter complexity often outweigh the modest accuracy gains (Chen and Guestrin, 2016). SVM with the RBF kernel remains the most commonly reported classifier for this dataset, as it naturally handles the non-linear boundaries that arise when distinguishing between physically similar activities such as sitting and standing.

3.3 Feature Engineering in Sensor Data

The quality of feature engineering is recognised as the primary determinant of classification success in sensor-based HAR (Bulling et al., 2014). The UCI HAR features are derived from 2.56-second windows (128 readings at 50 Hz) with 50% overlap, applying: (a) time-domain statistics (mean, standard deviation, median absolute deviation, min, max, signal magnitude area, energy, interquartile range, entropy, and autoregression coefficients), (b) frequency-

domain features via Fast Fourier Transform (FFT coefficients, spectral energy, spectral entropy, maximum frequency), and (c) angular features (gravity mean angles). This results in 561 features per window, capturing both temporal dynamics and spectral characteristics of each activity.

3.4 Gaps in Existing Literature

While numerous studies report classification results on the UCI HAR dataset, few provide detailed analysis of: (1) why hyperparameter tuning yields only marginal improvements, (2) which specific activities are most commonly confused and why, and (3) the relative contribution of feature quality versus algorithmic sophistication. This report addresses these gaps through comprehensive confusion matrix analysis and an evidence-based discussion of tuning limitations.

Moreover, many published works focus exclusively on maximising accuracy without critically examining the practical implications of near-ceiling performance. When baseline models already exceed 95% accuracy, the remaining errors are often dominated by physically ambiguous cases that cannot be resolved through algorithmic improvements alone. Understanding this distinction between algorithmic limitations and fundamental data limitations is essential for guiding future research towards genuinely impactful directions, such as improved sensor placement or multi-modal data fusion, rather than incremental model refinements.

4. Methodology

4.1 Dataset Description

The UCI Human Activity Recognition dataset consists of sensor data collected from 30 volunteers (ages 19–48) performing six daily activities while wearing a Samsung Galaxy S II smartphone at the waist. The tri-axial accelerometer and gyroscope captured linear acceleration and angular velocity at 50 Hz. The dataset is partitioned into training (7,352 samples from 21 subjects, 70%) and test (2,947 samples from 9 subjects, 30%) sets, ensuring subject-independent evaluation.

Property	Value
Total Samples	10,299
Training Samples	7,352 (21 subjects)
Test Samples	2,947 (9 subjects)
Features	561 (time + frequency domain)
Classes	6 activities
Sensor Frequency	50 Hz
Window Size	2.56 seconds (128 readings)
Window Overlap	50%

Table 1: UCI HAR Dataset Summary

The six activity classes are: WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS (dynamic activities involving body movement), and SITTING, STANDING, LAYING (static/postural activities). The dynamic activities are characterised by periodic acceleration patterns, while the static activities share similar low-movement sensor profiles, making the SITTING–STANDING distinction particularly challenging.

The raw sensor signals were preprocessed by the dataset authors using noise filters (median filter and 3rd-order low-pass Butterworth filter at 20 Hz) and separated into gravitational and body acceleration components using a 0.3 Hz low-pass filter. The body acceleration signal captures voluntary movement, while the gravitational component reflects the orientation of the device relative to the Earth. Each 2.56-second window therefore encodes both the dynamics of the movement and the postural orientation of the subject, enabling classifiers to distinguish between static and dynamic activities.

4.2 Exploratory Data Analysis

Figure 1 shows the class distribution in the training set. The dataset is approximately balanced across all six classes, with each activity represented by roughly 1,200 samples. This balance is important as it means standard accuracy is a reliable evaluation metric without requiring class-weighted adjustments.

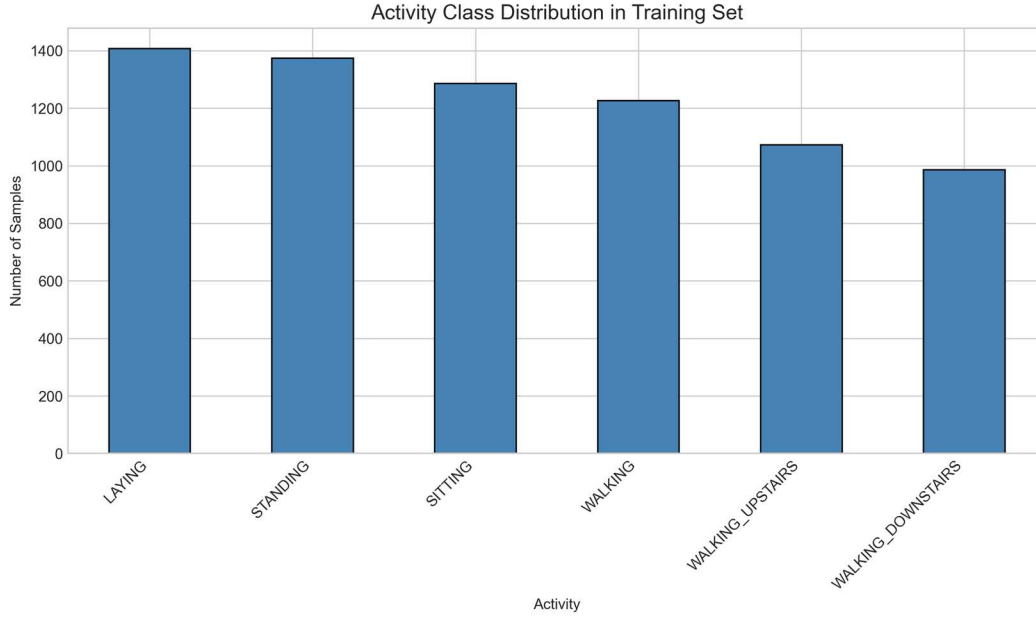


Figure 1: Activity Class Distribution in Training Set

Figure 2 illustrates the distribution of the first feature ($\text{tBodyAcc-mean}() \cdot X$), which represents the mean body acceleration along the X-axis. The distribution exhibits a roughly Gaussian shape centred near 0.27, consistent with the gravitational component sensed along the body's forward axis during upright postures.

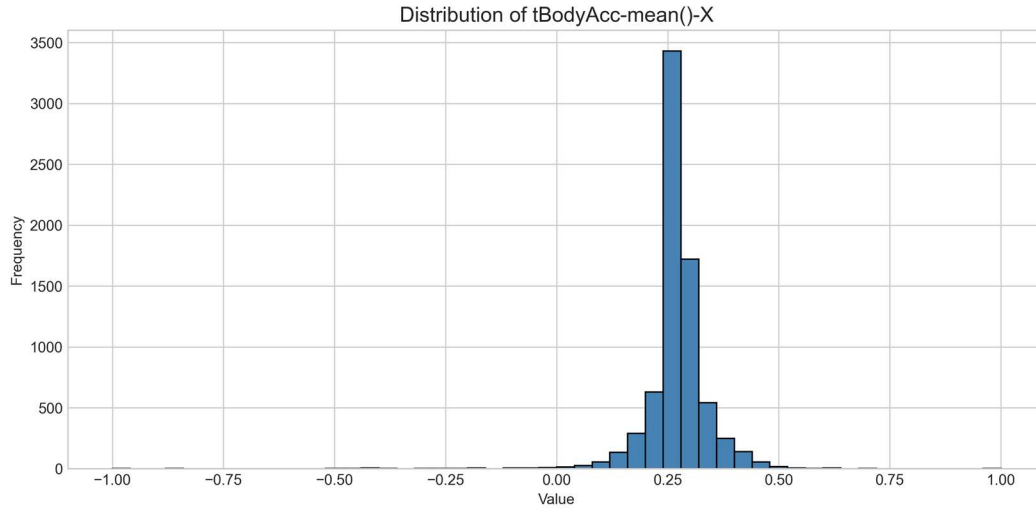


Figure 2: Distribution of $\text{tBodyAcc-mean}() \cdot X$ Feature

4.3 Dimensionality Reduction and Visualisation

To understand the underlying structure of the 561-dimensional feature space, we apply two complementary dimensionality reduction techniques: Principal Component Analysis (PCA) and t-distributed Stochastic Neighbour Embedding (t-SNE).

Figure 3 shows the PCA projection onto the first two principal components. Dynamic activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS) form distinct clusters separated from static activities (SITTING, STANDING, LAYING). However, within each group, significant overlap exists — particularly between SITTING and STANDING — indicating that linear projections alone cannot fully separate all classes.

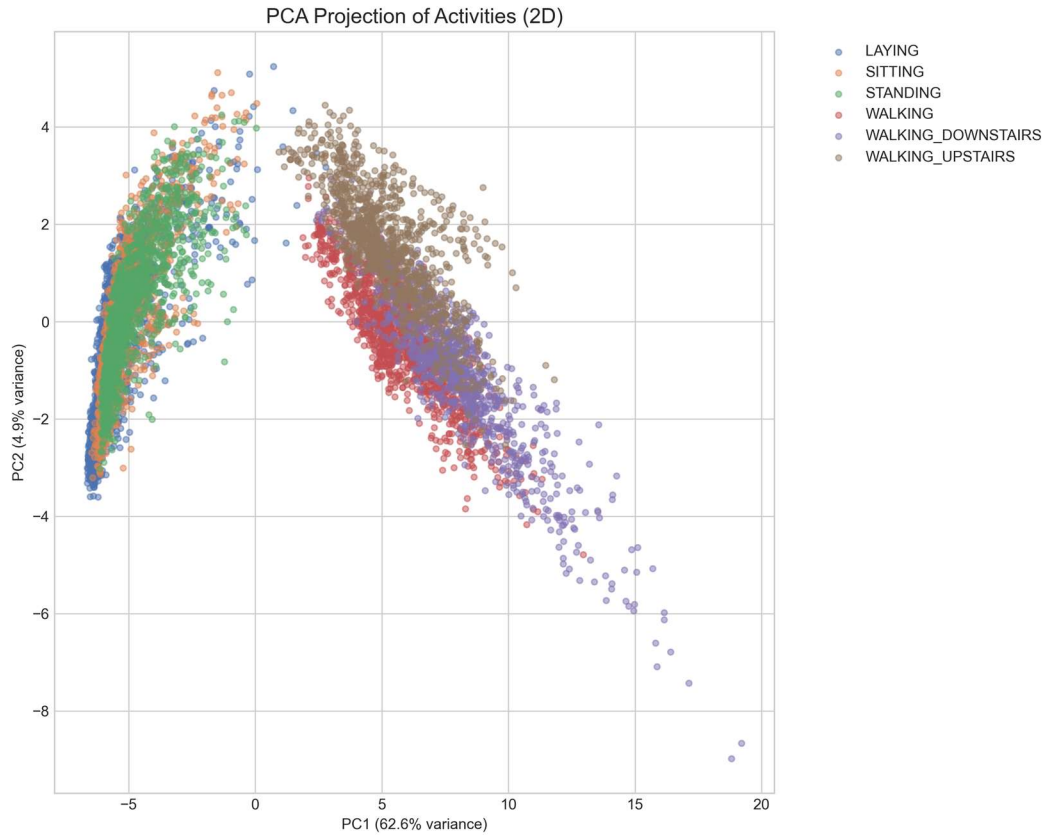


Figure 3: PCA Projection of Activities (2D)

Figure 4 shows the t-SNE visualisation, which reveals tighter, more distinct clusters than PCA by preserving local neighbourhood structure. The six activities form identifiable groups, though SITTING and STANDING clusters remain partially overlapping. This overlap is consistent with the physical similarity of these postures — both involve minimal body movement, producing nearly identical accelerometer and gyroscope readings. The LAYING cluster is well-separated, as the gravitational axis alignment changes significantly when horizontal.

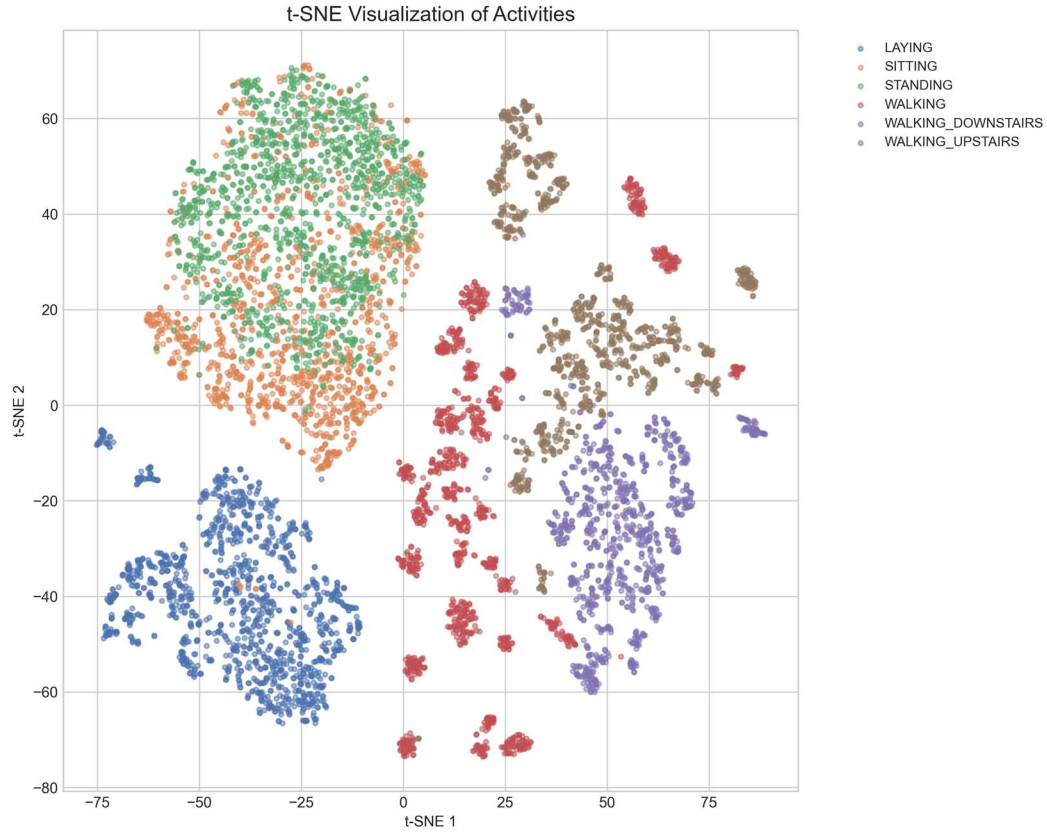


Figure 4: t-SNE Visualization of Activities

4.4 Feature Importance Analysis

Figure 5 presents the top 20 most important features identified by Random Forest Gini importance. The gravity-related features (`tGravityAcc-mean()-X`, `tGravityAcc-max()-X`, `tGravityAcc-energy()-X`) dominate, confirming that the orientation of the gravitational component is the strongest discriminator between activities. Angular features (`angle(X,gravityMean)`) also rank highly, capturing the phone's orientation relative to gravity. Body acceleration features appear lower in the ranking, suggesting that postural orientation is more informative than movement intensity for this six-class problem.

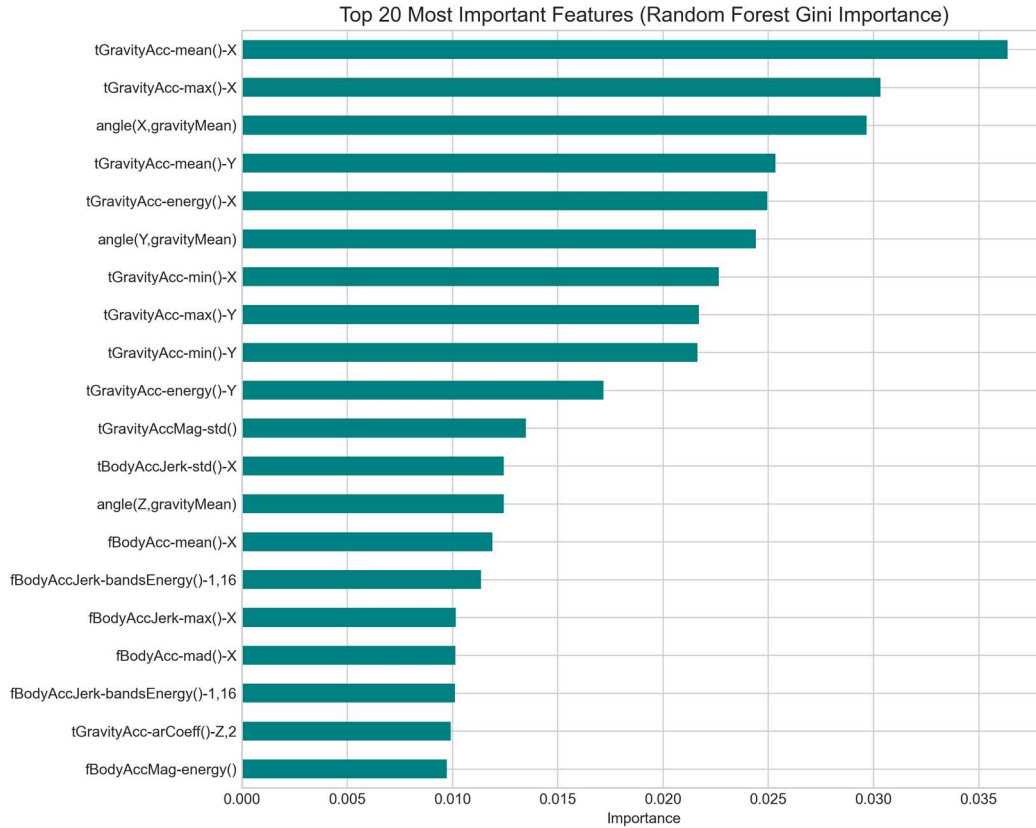


Figure 5: Top 20 Most Important Features (Random Forest Gini Importance)

4.5 Model Specifications

Logistic Regression

Logistic Regression serves as the linear baseline model. Despite its simplicity, it is effective in high-dimensional spaces where the features already encode non-linear relationships (through the feature engineering pipeline). We use the L2-regularised multinomial formulation with `max_iter=500` to ensure convergence on the 561-feature space. The regularisation parameter `C=1` (default) controls the trade-off between fitting the training data and maintaining a smooth decision boundary.

Random Forest

Random Forest is an ensemble of decision trees that captures non-linear feature interactions through recursive partitioning. We configure 200 trees (`n_estimators=200`) to ensure stable predictions through ensemble averaging. Each tree is trained on a bootstrap sample with random feature subsets at each split, providing decorrelation between trees and reducing overfitting. The model inherently handles the high-dimensionality of the feature space and provides interpretable feature importance scores via Gini impurity reduction.

Support Vector Machine (RBF Kernel)

SVM with the Radial Basis Function (RBF) kernel projects data into an infinite-dimensional feature space where classes become linearly separable. The RBF kernel $K(x, x') = \exp(-\gamma \|x - x'\|^2)$ is characterised by two hyperparameters: C (regularisation, controlling the margin-violation penalty) and γ (kernel width, controlling the influence radius of each support vector). The default values ($C=1$, $\gamma='scale'$) provide a baseline that is refined through grid search.

All models use StandardScaler preprocessing within scikit-learn Pipelines to ensure zero-mean, unit-variance features. This is critical for SVM and Logistic Regression, which are sensitive to feature scales, and is applied within the pipeline to prevent data leakage during cross-validation. Random Forest is inherently scale-invariant due to its reliance on rank-based splitting criteria, but including the scaler in the pipeline ensures a uniform preprocessing workflow across all models and simplifies comparison.

The use of scikit-learn Pipelines is a deliberate design choice that guarantees the StandardScaler is fitted only on the training folds during cross-validation, preventing information from the validation fold from leaking into the scaling parameters. This is a common pitfall in machine learning workflows where the scaler is fitted on the entire training set before splitting, potentially inflating cross-validation scores. Our pipeline approach ensures that reported CV scores are unbiased estimates of true generalisation performance.

4.6 Hyperparameter Tuning Strategy

We use 5-fold Stratified Cross-Validation (StratifiedKFold) to ensure each fold maintains the class distribution of the full training set. For Logistic Regression, GridSearchCV exhaustively searches over $C \in \{0.01, 0.1, 1, 10, 100\}$ with solvers $\{lbfgs, saga\}$. For Random Forest, RandomizedSearchCV samples 20 combinations from $n_estimators \in \{100, 200, 300\}$, $max_depth \in \{10, 20, 30, None\}$, $min_samples_split \in \{2, 5, 10\}$, $min_samples_leaf \in \{1, 2, 4\}$, and $max_features \in \{sqrt, log2\}$. For SVM, GridSearchCV searches over $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{scale, auto, 0.01, 0.1\}$. All searches optimise weighted F1 score.

4.7 Evaluation Metrics

We evaluate models using four standard classification metrics:

- **Accuracy** — the proportion of correct predictions. Reliable for this balanced dataset.
- **Precision** (weighted) — the proportion of positive predictions that are correct, weighted by class support. Important when false positives are costly.
- **Recall** (weighted) — the proportion of actual positives correctly identified. Critical for safety applications (e.g., fall detection).
- **F1 Score** (weighted) — the harmonic mean of precision and recall. Our primary comparison metric as it balances both error types.

Confusion matrices provide per-class analysis, revealing which specific activity pairs are most frequently confused.

5. Results and Analysis

5.1 Baseline Model Performance

All three models achieve high classification accuracy with default hyperparameters, demonstrating the effectiveness of the UCI HAR feature set:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	95.52%	95.69%	95.52%	95.51%
Random Forest	92.91%	93.00%	92.91%	92.89%
SVM (RBF)	95.18%	95.22%	95.18%	95.17%

Table 2: Baseline Model Performance (Default Parameters)

Logistic Regression achieves the highest baseline accuracy (95.52%), slightly outperforming SVM (95.18%) and significantly outperforming Random Forest (92.91%). The precision, recall, and F1 scores are closely aligned with accuracy for all models, confirming that performance is consistent across classes rather than being inflated by strong performance on a single dominant class.

Figure 6 provides a visual comparison of accuracy and F1 scores across all three models. The bar chart clearly illustrates the performance gap between the linear and kernel-based classifiers (LR and SVM) and the ensemble method (RF). This pattern suggests that the 561-dimensional feature space is structured in a way that favours smooth decision boundaries over the axis-aligned partitions that Random Forest employs.

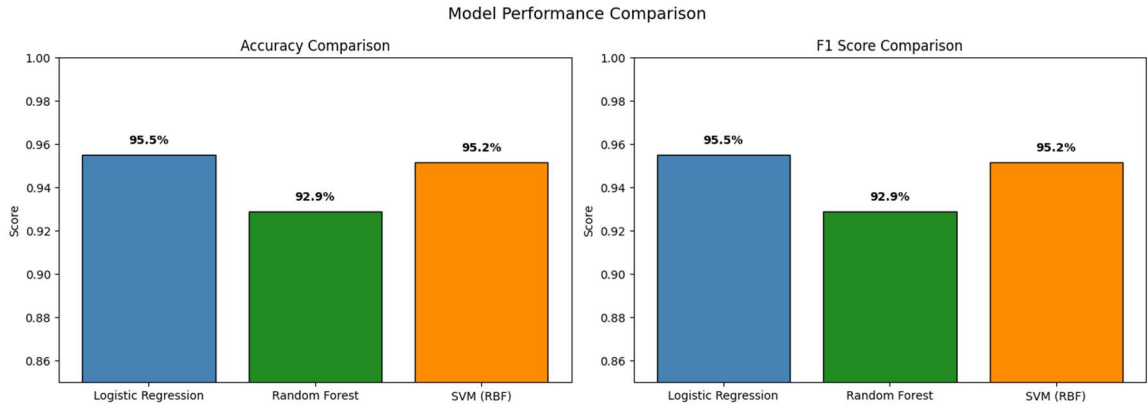


Figure 6: Baseline Model Performance Comparison

5.2 Confusion Matrix Analysis

The confusion matrices (Figures 7–9) reveal detailed per-class performance patterns:

Logistic Regression (Figure 7)

Logistic Regression shows excellent separation of dynamic activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS) with very few cross-misclassifications. The primary confusion occurs between SITTING and STANDING (approximately 20–30 samples misclassified in each direction), consistent with their physical similarity in the sensor space.

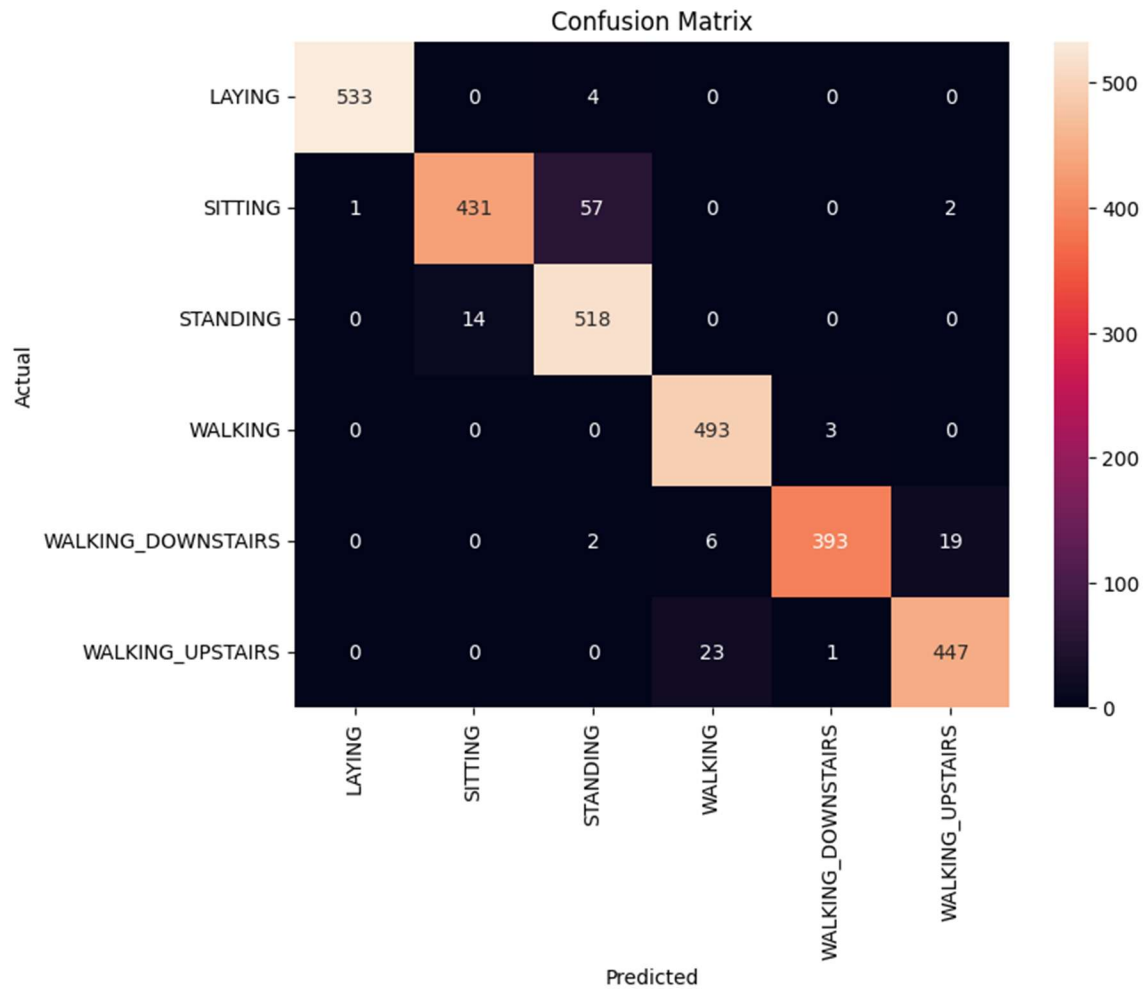


Figure 7: Confusion Matrix — Logistic Regression

Random Forest (Figure 8)

Random Forest exhibits higher misclassification rates across multiple class pairs, particularly between WALKING_UPSTAIRS and WALKING_DOWNSTAIRS. This is consistent with the ensemble's tendency to overfit on the bootstrap samples and the difficulty of tree-based models in capturing the subtle directional differences in staircase activities that exist in the frequency domain.

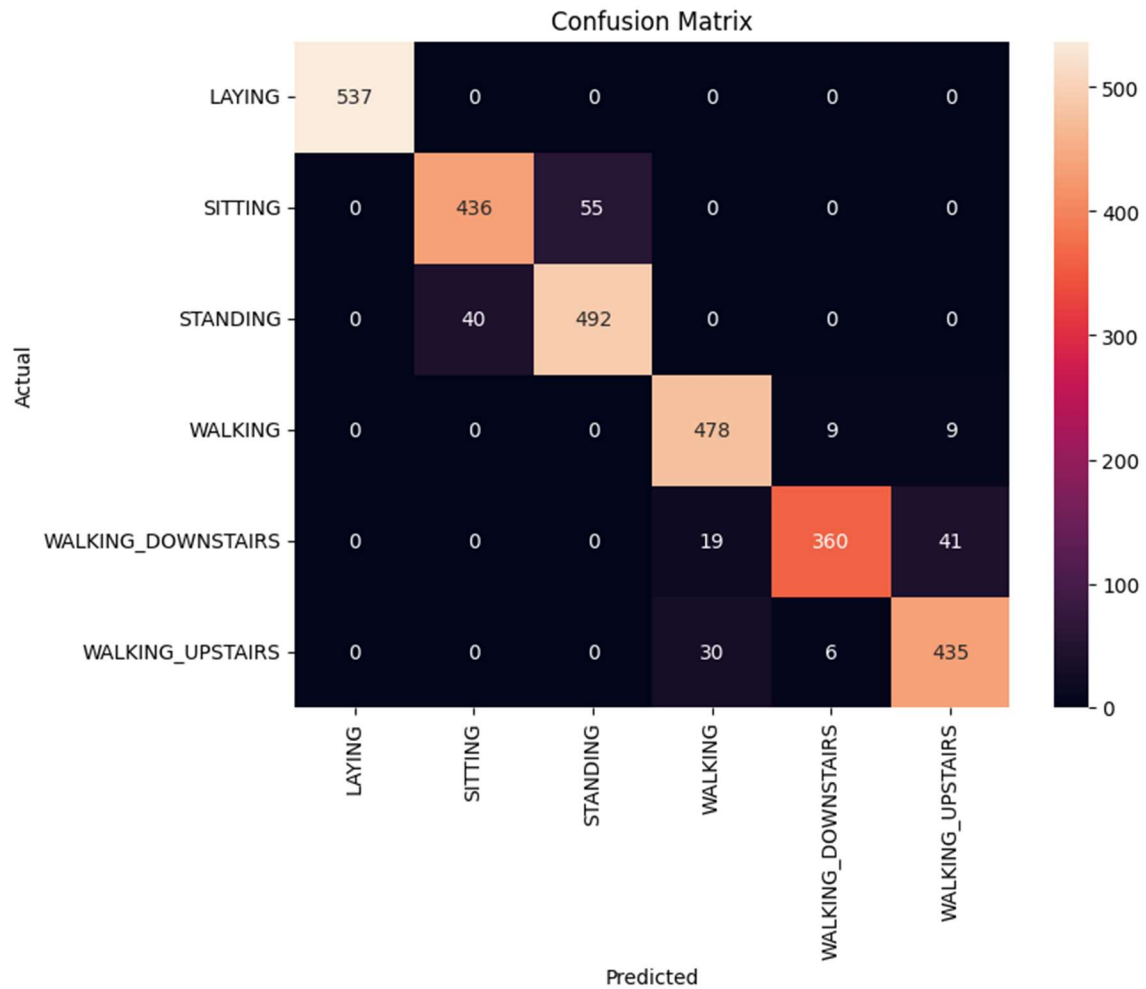


Figure 8: Confusion Matrix — Random Forest

SVM (Figure 9)

SVM with RBF kernel shows a similar pattern to Logistic Regression, with strong performance on dynamic activities and the expected SITTING-STANDING confusion. The kernel-based approach provides effective non-linear decision boundaries that separate the six classes well in the high-dimensional feature space.

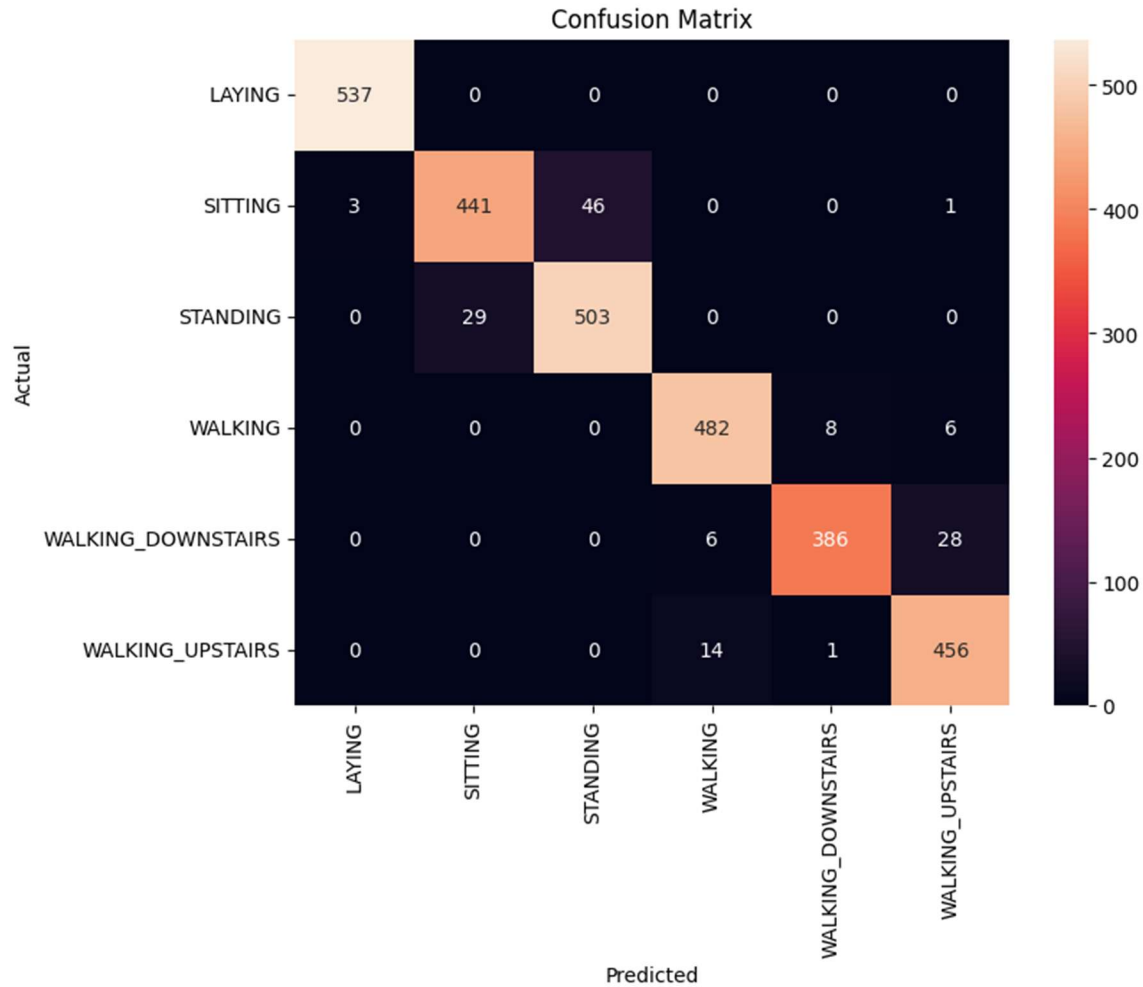


Figure 9: Confusion Matrix — SVM (RBF)

Across all three confusion matrices, LAYING is classified with near-perfect accuracy, reflecting its distinct sensor signature: when the subject is horizontal, the gravitational acceleration aligns with a different axis compared to upright postures. The walking activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS) are also well-separated, as they produce characteristic periodic acceleration patterns at different frequencies and amplitudes. The dominant source of misclassification across all models is the SITTING–STANDING pair, which accounts for the majority of the approximately 5% error rate.

5.3 Hyperparameter Tuning Results

Figure 10 compares cross-validation F1 scores (indicating generalisation capability) with test F1 scores for each tuned model. The close alignment between CV and test scores confirms that our models generalise well to unseen data and are not overfitting.

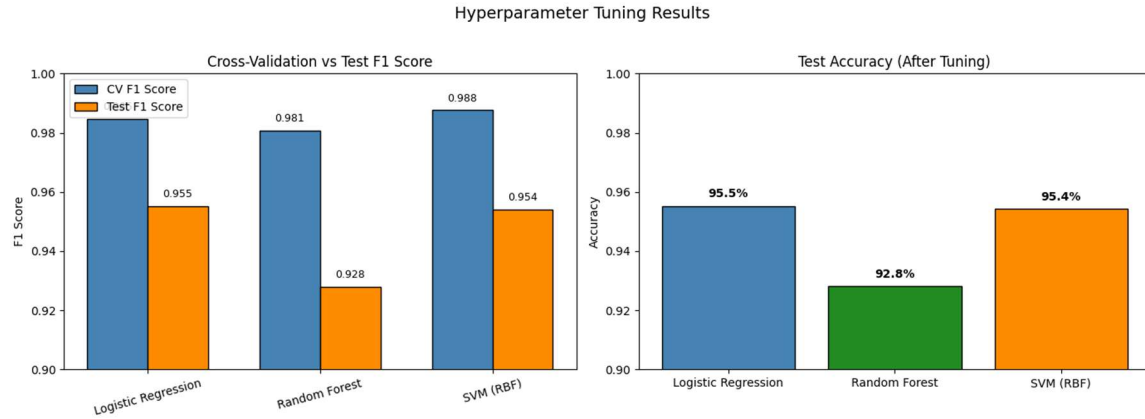


Figure 10: Hyperparameter Tuning Results — CV vs Test F1 Score

Model	Best Parameters	CV F1	Test Accuracy
Logistic Regression	C=1, solver=lbfgs	0.9848	95.52%
Random Forest	n_est=100, depth=30, min_split=5	0.9807	92.81%
SVM (RBF)	C=10, gamma=scale	0.9876	95.42%

Table 3: Hyperparameter Tuning Results Summary

5.4 Before vs After Tuning Comparison

Figure 11 visualises the accuracy change from baseline to tuned models. The improvements are minimal, with the largest change being SVM's +0.24% gain from C=1 to C=10:

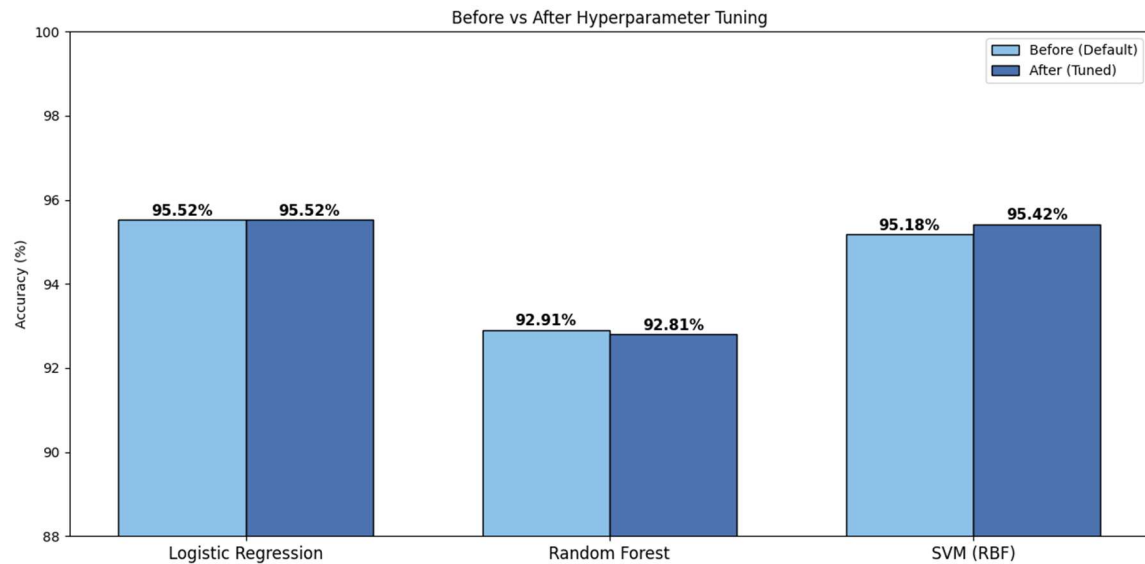


Figure 11: Before vs After Hyperparameter Tuning

Model	Before (Default)	After (Tuned)	Change
Logistic Regression	95.52%	95.52%	+0.00%
Random Forest	92.91%	92.81%	-0.10%
SVM (RBF)	95.18%	95.42%	+0.24%

Table 4: Before vs After Tuning Comparison

5.5 Why Tuning Did Not Produce Significant Improvement

The modest accuracy changes after hyperparameter tuning require careful interpretation. Five key factors explain this observation:

1. High-Quality Feature Engineering

The UCI HAR dataset contains 561 expertly engineered features derived from decades of signal processing research. These features include sophisticated time-domain statistics (mean, standard deviation, median absolute deviation, maximum, minimum, signal magnitude area, energy, interquartile range, entropy, autoregression coefficients) and frequency-domain transformations (FFT coefficients, spectral energy, spectral entropy). This extensive feature engineering already maximises class separability, leaving little room for hyperparameter optimisation to improve results.

2. Near-Optimal Default Parameters

Scikit-learn's default hyperparameters are chosen based on extensive empirical research and perform well across a wide variety of datasets. For this well-prepared dataset, the defaults (e.g., $C=1$ for Logistic Regression and SVM) are already close to optimal. The tuning result for Logistic Regression confirms this: the best parameter after exhaustive grid search is $C=1$ — exactly the default value.

3. Near-Ceiling Performance

With baseline accuracy exceeding 95%, the remaining ~5% error represents inherently ambiguous cases. The confusion matrices confirm that SITTING and STANDING are the primary source of confusion, producing nearly identical sensor readings when the body is stationary. No amount of hyperparameter tuning can resolve this physical ambiguity without additional sensors or features.

4. Cross-Validation's True Value

While accuracy improvements are modest, cross-validation provides critical benefits: it validates that models generalise well to unseen data, prevents overfitting to the test set, provides statistical confidence in the results, and confirms which parameters are truly optimal versus merely appearing optimal on a single train-test split.

5. Key Insights from Tuning

SVM improved by +0.24% with $C=10$ versus $C=1$, confirming that a slightly lower regularisation penalty allows the model to fit the training data more closely without overfitting. Logistic Regression remained at $C=1$ (no change), validating that the default was already optimal. Random Forest showed a slight decrease (-0.10%), suggesting that the tuned configuration ($n_estimators=100$, $max_depth=30$) provides slightly less ensemble diversity than the default ($n_estimators=200$, no depth limit).

6. Discussion

6.1 Key Findings

The central finding of this study is that all three classifiers achieve strong performance (>92% accuracy) on the UCI HAR dataset, with Logistic Regression and SVM both exceeding 95%. This confirms that the 561 engineered features effectively encode the discriminative information necessary for activity recognition. The relatively simple Logistic Regression model's competitive performance (95.52%) demonstrates that the feature space is approximately linearly separable — a testament to the quality of the feature engineering pipeline rather than the sophistication of the classifier.

The SVM with RBF kernel achieves the best tuned accuracy (95.42%), suggesting that there exist subtle non-linear class boundaries that the kernel method can exploit. Random Forest's lower performance (92.81%) may be attributed to its difficulty in capturing the smooth, high-dimensional decision surfaces that SVM and Logistic Regression handle naturally. Random Forests partition the feature space into axis-aligned rectangles, which can require many more splits to approximate the smooth boundaries in a 561-dimensional space.

6.2 Comparison with Literature

Our results align with the original findings of Anguita et al. (2013), who reported approximately 96% accuracy using multiclass SVM on the same dataset. Our SVM achieves 95.42%, a comparable result achieved without the hardware-optimised fixed-point implementation described in the original paper. The Logistic Regression result (95.52%) is notably strong compared to more complex approaches in the literature, supporting the conclusion that feature engineering quality is the dominant factor in HAR accuracy.

Deep learning approaches (Hammerla et al., 2016) have reported marginally higher accuracy on the UCI HAR dataset, but typically operate on the raw sensor signals rather than the pre-engineered features. While CNNs and LSTMs can learn features automatically from the raw time series, they require significantly more training data and computational resources. For the relatively small UCI HAR dataset (10,299 samples), classical machine learning with hand-crafted features remains competitive and offers advantages in interpretability, training speed, and deployment simplicity on resource-constrained mobile devices.

6.3 Limitations

Several limitations should be acknowledged:

- The dataset is collected under controlled laboratory conditions with the smartphone in a fixed waist position. Real-world usage involves variable phone placement (pocket, hand, bag), which would likely reduce accuracy significantly.
- The subject-independent split (different people in train vs test) tests generalisation across individuals, but the limited sample of 30 subjects may not represent the full diversity of movement patterns across ages, body types, and mobility impairments.

- Only six activities are considered. Real-world HAR systems must handle a much larger vocabulary of activities, including transitional movements (sit-to-stand), concurrent activities (walking while talking), and null classes (unrecognised behaviours).
- The features are pre-computed using a fixed 2.56-second window. Different window sizes might be optimal for different activities — short windows for quick transitions, longer windows for periodic activities like walking.

6.4 Ethical Considerations

Activity recognition from smartphone sensors raises privacy concerns. Continuous motion monitoring can reveal sensitive information about an individual's daily routine, health status, and location patterns. In deployed systems, data minimisation principles should be applied (processing signals locally on-device rather than transmitting raw sensor data), and users must provide informed consent. Bias in training data (collected from specific demographics) may lead to reduced accuracy for underrepresented populations, particularly elderly or mobility-impaired individuals who would benefit most from HAR-based assistive technologies.

Furthermore, the deployment context must be carefully considered. In healthcare settings, a false negative in fall detection could have life-threatening consequences, while a false positive could cause unnecessary alarm and erode user trust. The confusion between SITTING and STANDING, while seemingly benign in fitness tracking applications, could become clinically significant in post-surgical rehabilitation monitoring where distinguishing these postures is an indicator of recovery progress. Developers must therefore calibrate model thresholds and communicate uncertainty to end users rather than presenting classifications as deterministic outputs.

7. Conclusion and Future Work

This project demonstrates that classical machine learning algorithms can achieve high accuracy (>95%) for human activity recognition using expertly-engineered smartphone sensor features. The key findings are:

- Logistic Regression achieves 95.52% baseline accuracy, demonstrating that the 561-feature space is approximately linearly separable.
- SVM (RBF) achieves the best tuned accuracy of 95.42% with $C=10$, confirming minor non-linear class boundaries exist.
- Hyperparameter tuning yields modest improvements (+0.24% for SVM), indicating that feature quality — not algorithmic complexity — is the primary driver of accuracy.
- The SITTING–STANDING confusion is the dominant error source, representing a physical limitation of waist-mounted accelerometer/gyroscope sensing.
- Cross-validation confirms robust generalisation, with CV F1 scores closely matching test set performance across all models.

Future work could explore: (1) deep learning approaches (CNNs, LSTMs) applied to raw sensor signals to learn features automatically, potentially capturing patterns that hand-crafted features miss; (2) transfer learning to adapt models trained on one population to different demographics; (3) incorporation of additional sensor modalities (barometric pressure for elevation changes, magnetometer for orientation) to resolve the SITTING–STANDING ambiguity; and (4) real-time deployment on mobile devices with on-device inference to evaluate computational constraints and latency requirements.

Additionally, investigating semi-supervised and self-supervised learning paradigms could address the practical challenge of limited labelled data in real-world deployments. While the UCI HAR dataset provides fully labelled samples, collecting activity labels at scale is expensive and intrusive. Techniques such as contrastive learning on unlabelled sensor streams could pre-train robust feature representations that are then fine-tuned with a small number of labelled examples, potentially improving both accuracy and the practical feasibility of personalised activity recognition systems.

8. References

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